

MALNAD COLLEGE OF ENGINEERING, HASSAN
Fourth Semester (Autonomous Programme)
Semester End Examinations, June 2019

Course Title : Object Oriented Programming with Java Course Code : CS 403
Duration : 3 Hours Maximum Marks : 100

Instructions to the Candidates :

- i) Answer any FIVE full questions choosing atleast one from each part.
- ii) Missing data, if any, may suitably be assumed.
- iii) On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.

PART - A

1. a) Define OOP. Explain the salient features of object oriented programming. *CO1, P01, L2* 08
b) List and explain Java Buzz words. *CO1, P03, L2* 08
c) Briefly explain the Java development tool kit. *CO1, P03, L2* 04
2. a) Define class. Explain with an example to create an object of a class. *CO2, P01, L2* 04
b) Define constructor. Explain the different types of constructor with a code snippet. *CO2, P03, L2* 08
c) Write a Java program to find $y=1+x+x^2+x^3+....x^n$ using for - each loop construct. *CO2, P01, L2* 08

PART - B

3. a) Define access specifier. Explain the different access specifier with an example. *CO2, P01, L2* 08
b) What is function overloading? Write a Java program to find the area of a circle and triangle using function overloading. *CO2, P03, L2* 08
c) Define and explain the syntax of variable length arguments. *CO2, P01, L2* 04
4. a) Define inheritance. Explain the different types of inheritance. *CO2, P01, L2* 08
b) Define super () keyword. Explain how super () is useful in maintaining the order of calling the constructor. *CO2, P03, L2* 06
c) Briefly explain the function overriding with an example. *CO2, P03, L2* 06

PART - C

5. a) Define interface. Explain with a suitable example to create an interface. *CO3, P03, L2* 06
b) Explain how multiple inheritance can be achieved using interfaces. *CO3, P01, L2* 08
c) Briefly explain the static import with a suitable example. *CO3, P03, L2* 06
6. a) Define exception. Explain the exception handling mechanism. *CO3, P01, L2* 06
b) Write a Java program to demonstrate *CO3, P03, L2* 08
(i) ArrayIndexOutOfBoundsException. (ii) Arithmetic exception.
c) Explain (i) throw (ii) throws (iii) finally *CO3, P01, L2* 06

PART - D

7. a) What is a thread? Explain the different ways of creating a thread. *CO4, P02, L2* 09
b) List the different synchronization methods. *CO4, P03, L2* 02
c) Write a Java program to create two threads A&B- which performs 09
i) Thread A: Displays the numbers from 1 to 50
ii) Thread B: Displays "MCE HASSAN" for 50 times. Demonstrate method synchronization. *CO4, P03, L2*

P.T.O.

approach	IoT based waste management for smart cities	Drone deep reinforcement Learning: A Review	artificial intelligence
Bank fraud detection using community detection algorithm	Evolution of Darknet analysis and Industrial control system usage	Drone deep reinforcement Learning: A Review	artificial intelligence

8. a) Define Java I/o. List any four I/o classes:

CO4, P02, L2

b) Write a Java program to create and display the name and path of a file. CO4, P03, L2

c) What are Byte streams? Write a Java program to copy the contents of one file to another file.

CO4, P03, L2

Course Code : CS 504

Maximum Marks : 100

Instructions to the Candidates :
 2. Answer any FIVE full questions choosing atleast one from each part.

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PART – A

1. a) Write a program in java to display invert triangle. For example: input : 4 output : 44444
333
22
1. 06
- b) List the primitive data types in java. With suitable example, describe any two of them. 06
- c) Define a java class to hold a list containing USN, Name and marks in 3 subjects. Add constructor and a method to compute average marks. Write a demo class to call the method. 08
2. a) Write the syntax of for each loop. Demonstrate how it can be used for printing a two dimensional array. 06
- b) Exemplify with a program how "this" keyword resolves namespace collision in java? 06
- c) List the looping constructs in java. Use them to write a program in java to find the count of and sum of all integers greater than 100 and less than 200 that are divisible by 7. 08

PART - 2

- PART - II

3. a) Define recursive method.? Write a recursive method that displays the contents of the string backwards. 06

b) Describe the purpose and use of both versions of "super" keyword. 06

c) Write a java program to compute the area of a triangle, rectangle and square using the abstract class. 08

4. a) Write a java program to check if the given input sentence is a pangram or not, using Inheritance. (pangram is a sentence which contains all the letters of an alphabet at least once.). 06

b) Write a local class Bitout within a main method to print all the integer values 0 to 9 in Binary format. 06

c) Define Inheritance? Demonstrate the use of overloaded constructors in Multilevel Inheritance. 08

PART -C

- PART -C

5. a) Mention any two Java's standard package? Write a java program to create a package called "student" and import it. 10

b) Define Exception. List different clauses of Exception. With an example, explain any two exception handlers. 10

6. a) Define constants and variables in interface? With an example explain nested interface. 10

b) Compare and contrast throws and throw exception handlers used in Java run time system. 10

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Note: Revealing of identity in any form in the answer book will be treated as malpractice.

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CS017	IoT based waste management for smart cities
8CS035	Detect and classify zero day malware
18CS010	Drone deep Reinforcement Learning: A Review

PART -D

7. a) What is Multithreading? Write a program to create multithreads in JAVA.
b) Discuss the significance of synchronization in JAVA?
8. a) List Applet initialization and termination methods. Write a java applet that sets the background color to cyan and foreground color to red and output a string message "A simple Applet".
b) What are applets? Exemplify different stages in the cycle of an applet along with its skeleton.

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MALNAD COLLEGE OF ENGINEERING, HASSAN

Sixth Semester (Autonomous Programme)
Supplementary Semester End Examinations, July - August, 2015

Course Title : Programming Using JAVA
Duration : 3 Hours

Course Code : CS 602

Maximum Marks : 100

Instructions to the Candidates :

Answer any FIVE full questions choosing atleast one from each part.
Missing data, if any, may suitably be assumed.

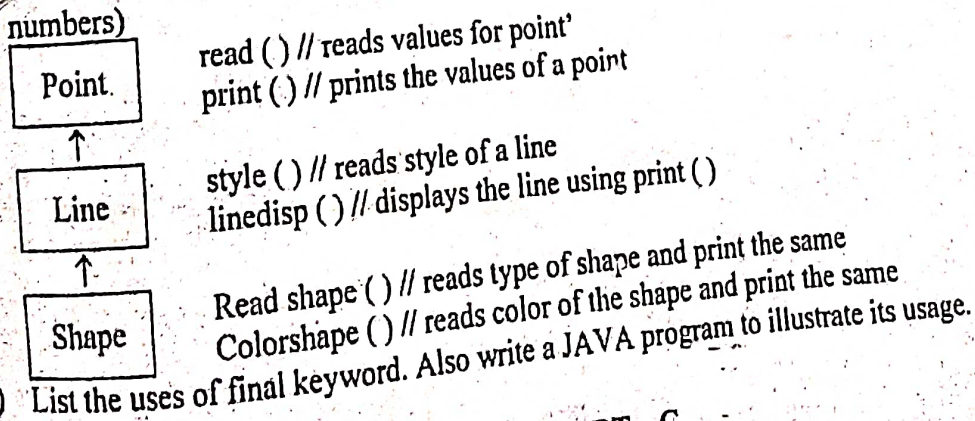
On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.

PART - A

- a) Explain Java's contribution to internet. 06
- b) Explain type conversion and casting in Java with suitable examples. 06
- c) What are Literals in Java? Explain different types of Literals. Give example for each. 08
- d) Write a Java program to add two rational numbers. (use returning value from a method) and display their result. 10
- e) Write a Java program to compute area of a circle and a rectangle using parameterized constructors. 10

PART - B

- a) What do you understand static in JAVA. List the restrictions on static. Write a Java program to demonstrate the use of static members and methods. 10
- b) Differentiate between method overloading and method overriding. Write a JAVA program to illustrate the use of method overloading and constructor overloading. (Should be implemented in a single program). 10
- a) Write a JAVA program to create a multilevel hierarchy for the given figure. (Assume suitable numbers) 12



- b) List the uses of final keyword. Also write a JAVA program to illustrate its usage. 08

PART - C

- a) Write a JAVA program to demonstrate the statement "Interfaces can be extended" and explain the same. 10
- b) Write a JAVA program to create static import in package with suitable example. Also explain. 10
- a) What are exceptions? Write a JAVA program to illustrate multiple catch clauses and finally. 10

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MALNAD COLLEGE OF ENGINEERING, HASSAN

Sixth Semester (Autonomous Programme)
 Makeup Semester End Examinations, June 2016

Course Title : Java Programming
 Duration : 3 Hours

Course Code : CS 602

Maximum Marks : 100

Instructions to the Candidates :

- Answer any FIVE full questions choosing atleast one from each part.
- Missing data, if any, may suitably be assumed.
- On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.

PART - A

- List and explain primitive data types used in Java. Give examples. 10
 - Write a Java program to find the sum and average of even numbers within a given range. 10
- With the help of a general form, explain the working of switch statement. Give example. 06
 - What is type casting in Java? Explain with a suitable example. 04
 - Write a Java program to find the sum of digits of a given number (eg: Num: 1524 Sum = 1+5+2+4=12) 10

PART - B

- Explain the following string handling functions
 i) length () (ii) indexOf () (iii) charAt () (iv) compareTo () (v) equals () 10
 - Write a Java program to find the sum of two matrices and print the resultant matrix in a matrix format. 10
- With a suitable example, explain overloading and overriding. 10
 - Explain the uses of super, static and final keywords. Give examples. 10

PART - C

- What is an interface? With a suitable example, explain how interface can be declared and inherited. 10
 - Explain the following access specifiers with reference to packages. Give suitable example
 i) Private (ii) Default (iii) Protected (iv) Public. 10
- What is an exception? Write a Java program to create your own exception. 10
 - Discuss the keywords try, catch, throw, finally and throws used in exception handling. 10

PART - D

- With a suitable example, explain the different ways to create a child thread. 10
 - With a suitable example, explain synchronization in multithreaded program. 10
- With the help of an applet skeleton, explain the life cycle of an applet. 10
 - Write a swing application program that has 2 buttons named "Alpha" & "Beta". When either of the buttons pressed, it should display "Alpha is pressed" and "Beta is pressed" respectively. 10

MALNAD COLLEGE OF ENGINEERING, HASSAN

Sixth Semester (Autonomous Programme)
Semester End Examinations, May / June, 2015

Course Title : Programming Using Java
Duration : 3 Hours

Course Code : CS 602
Maximum Marks : 100

Instructions to the Candidates :

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PART - A

- Briefly outline Java's contribution to inter 05
- Write the difference between normal and short circuit operators. With an example program demonstrate the short circuit AND operator. 05
- Write a java program to read a character from keyboard and display it. 05
- With a sample program illustrate how break can be used as a form of goto. 05
- Create a swapper class with two integer instance variables x and y. Include constructor and accessor methods. Also include a method called swap that exchanges the value of X and Y. Then create a demo class that checks all the methods. 07
- How does java support Garbage collection? 05
- Write a java program that finds the minimum and maximum value in a 3 x 3 matrix. The program should output the value and position of minimum and maximum element. 08

PART - B

- What is constructor overloading? Create a class called TwoDshape with appropriate instance variables. Define atleast three forms of constructor. Create objects by calling the constructors. 07
- With examples demonstrate the use of keyword static with variable declaration and method declaration. 06
- Create a varargs method called sum that adds the integer values passed to it. Have it return the result. Write a demo class to use it. 07
- Write a class called square with a method to calculate area. Create a subclass of square called cube with an overriding method that computer surface area of cube. Include a demo class to create a cube object. 08
- What is an abstract class? Write an abstract class with an abstract method void compute (int n). Implement this method in a subclass to find perfect squares from 1 to n. 08
- Illustrate atleast two uses of the keyword final. 04

PART - C

- How is an interface different from abstract class? Create an interface called data - structure and write an implementing class stack for data - structure. 10
- Write atleast two objective of using packages. With an example program of your choice demonstrate how to create a package in java. 10

MALNAD COLLEGE OF ENGINEERING, HASSAN

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PART - A

- Explain the three main principles of object-oriented programming. 06
 - Write a Java program to find the biggest of n numbers (without using arrays). 08
 - Explain any two iterative statements used in Java. Give suitable example. 06
- Why constructors are required? With a suitable example, explain parameterized constructors and constructor overloading. 06
 - Write a Java program to find the number of multiples of 2's in a given nxm matrix. 08
 - What is the value for x, y & z if a = 10 and b = 12 06
x = a & b;
y = a >>> b;
z = a << b;

PART - B

- With a suitable example, explain method overloading and method overriding. 08
 - Explain the following with a suitable example 06
i) static (ii) final
 - Explain nested and inner class. Give suitable example. 06
- Implement a method addAtEnd () that takes an integer array "data" and an integer 'x' as its parameters. It creates a new array whose length is one greater than data's length. It then copies all of data's elements into the new array and lastly add the value of x into the last elements of the array. It returns the new array. Write a Java program to add an element to an array by using addAtEnd (). 10
 - What is abstract class? Illustrate with a suitable example. 06
 - Explain the use of super. 04

PART - C

- What is an interface? With the help of general syntax, explain how to declare and inherit an interface. 08
 - Explain the different access specifiers used in Java. 08
 - What is static import? 04
- Explain the following keywords used in exception handling. 10
i) try (ii) catch (iii) throw (iv) throws (v) finally.

P.T.O.

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b) Explain the following with a suitable example 06
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4. a) Implement a method addAtEnd () that takes an integer array "data" and an integer 'x' as its parameters. It creates a new array whose length is one greater than data's length. It then copies all of data's elements into the new array and lastly add the value of x into the last elements of the array. It returns the new array. Write a Java program to add an element to an array by using addAtEnd (). 10
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PART - C

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P.T.O.

- b) Write a Java program to create a... throw your own exception.

PART - D

7. a) With a suitable example, explain how to create a child thread using Runnable interface.
b) What are applets? Explain the different stages in the life cycle of an applet.
c) Explain JButton and JLabel.
8. a) Write a Java program to solve producer-consumer problem.
b) Write an applet to scroll a message. The message has to be passed from an html file to the applet.

- b) Write a Java program to create a  throw your own exception.

PART - D

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